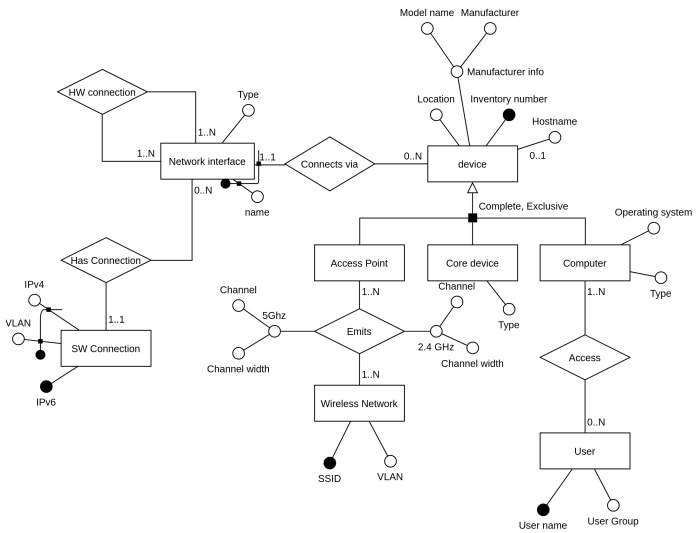

Figure 1: ER diagram

We have **devices** which are identified by their *inv number* and every device has it's *model name* and *Manufacturer*. It is located in a room (*location*) (which has it's own number and is uniquely identified by it) and if connected to network a *host-name*. Devices fall into several groups. **Core devices** such as switches and routers which do not have anything special but *type* (switch, router, etc.). **Access Points** where we need what **Wireless networks** it emits and over which channels and their parameters (*2.4 GHz* and *5 GHz channel* and it's *width*). These networks have some SSID and VLAN ID to which it belongs. Last type of device is **Computer** which has it's *type* (Laptop, All In One, Desktop, etc.) and *Operating system*. These computers can be accessed by different **users**, like teachers, students etc. which are all *user groups*. But not every teacher has access to all teacher devices (for example take-home laptops).

All **devices** connect somewhere via **Interface**. This interface identified by it's *name* scoped to device can be of different *type*. Mainly wireless or ethernet, but SFP, SFP+ and QSFP+ are all options. Every **interface** connects by wire or em wave to other **interfaces**. But these HW connections can carry multiple **software connections**. Which are identified by *IPv6* (which is unique) or *IPv4* which is unique only for it's *VLAN*.